

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 09 · LECTURE NOTES

Protein Structure

The problem that changed, and the parts that did not.

MODULE OUTCOMES

- State the folding problem and why it was hard.
- Describe the classical approaches and their limits.
- Explain what learned predictors changed.
- State honestly what remains unsolved.
- Interpret a predicted structure and its confidence.

1 The problem

THE SEQUENCE DETERMINES THE STRUCTURE, AND THE MAPPING WAS NOT COMPUTABLE

Anfinsen's experiment showed that a denatured protein refolds to the same structure without assistance — so the information required is present in the sequence, which made prediction a well-posed goal rather than a hope, and is why the problem was stated so confidently so early.

And the search space is astronomical — Levinthal's point: a random search over conformations would take longer than the age of the universe, and yet folding takes milliseconds — so the energy landscape must be funnelled toward the native state rather than flat.

Physical simulation from first principles was attempted for decades, with enormous computational investment — and the force fields were not accurate enough to remain correct over the required timescales, with small errors accumulating, which is an honest summary of why it did not work.

So the field split in practice: homology modelling where a related structure had been solved experimentally, and threading or fragment assembly where one had not — both of which used the database of known structures as the actual source of information, rather than physics.

2 What changed

LEARNED PREDICTORS REACHED EXPERIMENTAL ACCURACY FOR MANY PROTEINS, USING COEVOLUTION AS THE KEY SIGNAL

Residues that are in physical contact coevolve — a destabilising substitution at one position is compensated by a substitution at its partner — so a sufficiently deep alignment of a protein family contains information about which residues touch, which is Module 08 §4's covariation argument in three dimensions.

Extracting that signal required disentangling direct contacts from indirect correlations (A correlates with C merely because both contact B) — which was the key methodological step, and it produced usable contact predictions before deep learning was applied, a piece of history worth keeping.

And the deep models combined that coevolutionary signal with learned structural priors and end-to-end training against known structures — reaching accuracy comparable to experimental determination for a large fraction of targets in blind assessment, which is the strongest possible form of the claim.

Which is a genuine and large result — one of the clearest successes of machine learning on a scientific problem — and it is a result about structure prediction, which is narrower than 'the folding problem is solved' (§3 lists what it does not cover).

3 What remains

- **The folding process** — the predictors produce a final structure without simulating how it is reached — so **misfolding, folding kinetics, chaperone dependence, and aggregation are untouched**, and those are where several diseases are.
- **Proteins without deep alignments** — accuracy depends substantially on the depth of the available **family alignment**, since that is where the coevolutionary signal comes from — so **orphan sequences, rapidly evolving proteins, and designed proteins are considerably harder**.
- **Intrinsically disordered regions**, which have no single native structure at all — **and the predictors flag them with low confidence, which is exactly the right behaviour and is frequently ignored by the people using the output (§4)**.
- **Complexes, conformational change, and ligand-bound states** — a protein's function frequently *is* its motion between states, and a single static prediction cannot express that however accurate it is.
- **And function** — a structure is not a mechanism, and predicting the fold does not predict what the protein does (Module 03 §4's chain: structure is one more link, not the destination). **The distance from a predicted fold to a biological claim is the same distance this course has been measuring throughout.**

4 Reading a prediction

A PREDICTED STRUCTURE COMES WITH PER-RESIDUE CONFIDENCE, AND USING IT WITHOUT THE CONFIDENCE IS THE ERROR

The predictors emit a per-residue confidence score alongside the coordinates — and **low-confidence regions are frequently genuinely disordered rather than merely uncertain, which makes the score information rather than a disclaimer** and is a point worth making to anyone using the output.

So a figure showing the whole predicted structure at equal visual weight is misleading — colour by confidence, which the standard viewers do by default — and this is **CSCE 679 Module 09's uncertainty requirement** in a field where the uncertainty is supplied for free and discarded anyway.

And predicted structures are now deposited in public databases alongside experimentally determined ones — so the predicted-versus-measured distinction has to travel with the identifier, which is **Module 07 §4's predicted-annotation problem at a much larger scale** and with the same mechanism of loss.

Which is this module's version of the program's rule: 'the predicted structure, at this confidence, from this model version' — three things, and a citation in a paper usually gives none of them.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Anfinsen — Principles that govern the folding of protein chains	§1 — the experiment that made the problem well posed.
Marks et al. — Protein structure prediction from sequence variation (free)	§2's coevolution signal , before the deep models — and it is the conceptual core of what followed.
Jumper et al. — Highly accurate protein structure prediction with AlphaFold (free)	§2 — the method paper, and its own limitations section is the model for §3.
The CASP assessments (free)	§§2 and 3 — blind assessment over three decades, which is how the claim was established and is the right evidence to cite.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State Anfinsen's result** and what it licensed.
- 2 **State Levinthal's paradox** and its resolution.
- 3 **Explain why physical simulation did not succeed.**
- 4 **Explain coevolution as a contact signal** in four sentences.
- 5 **Explain direct versus indirect correlation** with a three-residue example.
- 6 **Fetch a predicted structure** and colour it by confidence.
- 7 **Find a low-confidence region** and check whether it is predicted disordered.
- 8 **Find a protein with a shallow alignment** and compare its prediction quality.
- 9 **List five things a predicted structure does not tell you.**
- 10 **Write a citation** that names the structure, the confidence, and the model version.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What did Anfinsen show, and why did it matter?
- 2 State Levinthal's paradox and what it implies about the landscape.
- 3 Why did first-principles simulation fail?
- 4 What did the field do instead?
- 5 What is the coevolution signal, and what had to be disentangled?
- 6 State the result carefully, and what it is narrower than.

- 7 Name five things that remain open.
- 8 Why is low confidence information rather than a disclaimer?
- 9 What should a figure of a prediction do?
- 10 What three things should a citation name?